Andrew Marion

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EDUCATION

KENNESAW STATE UNIVERSITY (KSU) | Marietta, GA — Anticipated Graduation May 2027

- *Bachelor of Science in Mechanical Engineering, Aerospace Engineering Minor* — GPA: **3.81/4.0**
- Relevant Coursework: Aerodynamics, Aircraft Design and Performance, Fluid Dynamics, Thermodynamics, Heat Transfer, Dynamics, Machine Design, Material Science, Computer Aided Engineering

RESEARCH EXPERIENCE

KSU Advanced Aerodynamics and Propulsion Lab | Marietta, GA — September 2023 – Present

Undergraduate Research Assistant, Advisor: Dr. Gaurav Sharma

Aerodynamic Characterization of a Blended Wing Body Using FlightStream — January 2026 – Present

- Benchmarked Siemens FlightStream against Spalart-Allmaras RANS CFD for the BlendedNet++ Geometry 227 blended wing body (NACA 25118, NASA SC(2)-0412, and NACA 0012 sections) in collaboration with Siemens
- Ran a 42-case transonic RANS sweep in ANSYS Fluent (Mach 0.6–0.8, -5° to 20° angle of attack)
- Showed FlightStream solved matched cases $5\times$ faster than RANS with lift coefficient within 0.013 in the linear range, supporting faster early-stage design iteration
- First author on 2 papers accepted to AIAA SciTech 2027; co-author on a Lattice Boltzmann analysis of the same aircraft

Control Surface Validation of the "Bird" UAV with FlightStream — March 2025 – Present

- Validated FlightStream aileron-deflection load predictions on a Group-1 fixed-wing UAV (S2F-BD213) against high-fidelity CFD and WindShape wind tunnel testing with a 6-axis load cell; published at AIAA SciTech 2026
- Built aerodynamic models that set initial flight conditions for closed-loop roll-axis control validation on a Pixhawk Cube Orange+ flight controller
- Cross-validated a mid-fidelity fixed-wing aerodynamic model against flight test data for control law development; accepted to AIAA SciTech 2027
- Coordinate data sharing and technical deliverables across a multi-institutional partnership with Oklahoma State University's S2FAR Lab, Siemens, and WindShape

Numerical Investigation of Compound Delta Wings — September 2023 – Present

- Executed 140+ CFD simulations in ANSYS Fluent and OpenFOAM across velocities and angles of attack, increasing lift-to-drag ratio 7.24–13.57% over the baseline wing in the transonic regime
- Studied vortex lift through trailing-edge boundary condition treatment and modeled laminar flow at low Reynolds numbers
- Applied GPU-accelerated Large Eddy Simulation (LES) to resolve vortex breakdown and validate RANS predictions
- Authored and presented first peer-reviewed paper at the 2025 AIAA Region II Student Conference

Tilt-Wing Propeller Modeling with Experimental Validation — October 2026 – Present

- Model an APC 16×10E propeller for tilt-wing and tilt-prop VTOL UAVs in Simcenter FlightStream by rotating the blade model with the unsteady surface-vorticity solver, compared against a blade element model with Pitt–Peters dynamic inflow
- Validating against six-component force-balance data from the WindShape facility at OSU–Tulsa across 0°–180° thrust inclination in 15° steps at 3000, 4000, and 5000 RPM
- Verified the reconstructed blade geometry against a 3D scan of the test propeller; running time step, mesh, and wake sensitivity studies before the full test matrix

PROFESSIONAL EXPERIENCE

NASA Glenn Research Center | Cleveland, OH — August 2026 – December 2026

UAV Flight Control Intern, Diagnostics and Controls Branch

- Implement MATLAB/Simulink flight control algorithms on Pixhawk flight computers running PX4 for a 12 ft wingspan distributed electric propulsion UAV with 16 electric fans and thrust vectoring, based on NASA's SUSAN Electrofan
- Develop fault-tolerant control reallocation algorithms in MATLAB that redistribute thrust commands after fan or thrust-vectoring failures, tested across 8,000 Monte Carlo fan-failure cases
- Perform system identification on flight test data using Kalman filtering and airspeed, IMU, and GPS sensor fusion
- Model UAV flight dynamics and powertrain behavior in MATLAB/Simulink and X-Plane, and integrate algorithms into PX4 in a Linux environment

Textron GSE | Cartersville, GA — June 2026 – August 2026

Mechanical Design Engineer Intern, Ground Support Equipment Electrification

- Designed 3 assemblies and 15 parts in Siemens NX with build-ready drawings, including the MX4 electric baggage tractor hinged door and 660 belt loader access doors
- Applied GD&T and tolerance stack-up analysis with DFA/DFM reviews to validate manufacturability before release

- Performed hands-on sheet metal design and prototyping, fit checks, and physical tests, including parking brake sensor location testing and access door motion verification

KSU Office of Research | Marietta, GA February 2026 – June 2026

Research Development Intern

- Conducted funding searches and budget analyses to identify external grant opportunities for KSU faculty
- Built NIH R15 and R16 grant resources, including proposal checklists and SharePoint pages
- Designed and built web pages communicating funding opportunities and research resources to faculty

TEACHING EXPERIENCE

Kennesaw State University | Marietta, GA September 2025 – January 2026

Dynamics Teaching Assistant

- Graded and annotated homework for 80+ students, explaining errors to reinforce dynamics fundamentals
- Held one-on-one sessions with students on dynamics concepts and problem solving
- Supported students over email and Microsoft Teams with homework questions

PUBLICATIONS

Conference Papers

- Marion, A., Sharma, G., Sirak, E., and Shahjahan, S., "Aerodynamic Characterization of a Blended Wing Body Using a Surface Vorticity Solver," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Marion, A., Sharma, G., Sirak, E., and Shahjahan, S., "Modeling Vortex-Dominated Flow over a Delta Wing Using a Surface Vorticity Solver," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Mehar, P., Marion, A., Sharma, G., Muller, M., and Gaedtke, M., "Aerodynamic Analysis of a Blended Wing Body Using the Lattice Boltzmann Approach," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Atkinson, Z., Comer, A., Marion, A., and Sharma, G., "Cross-Validation of a Mid-Fidelity Fixed-Wing Aerodynamic Model for Control Law Development with Flight Testing," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Atkinson, Z., Comer, A., Marion, A., Sharma, G., Shahjahan, S., and Wilson, T., "Multi-Fidelity Validation of Control Surface Deflection Predictions Using FlightStream," AIAA SciTech 2026 Forum, January 2026. doi.org/10.2514/6.2026-2810
- Marion, A., and Sharma, G., "Numerical Study of Transonic Flow Characteristics Around a Compound Delta Wing," AIAA Region II Student Conference, April 2025. doi.org/10.2514/6.2025-97857

Presentations, Posters, and Media

- Marion, A., "Study of Compound Delta Wings and Kennesaw State Undergraduate Research Experience," Invited presentation to the KSU President's Community Advisory Board, May 2025
- Marion, A., and Sharma, G., "Numerical Analysis of Vortex Breakdown Over a Compound Delta Wing," Poster, KSU Symposium of Student Scholars, May 2024
- Featured in Kennesaw State University News, "Engineering Student Uses Computer Simulations to Shape the Future of High-Speed Flights," March 2026. kennesaw.edu/news/stories/2026/engineering-student-computer-high-speed-flights.php

LEADERSHIP AND COMPETITION TEAMS

Kennesaw Rocketry Team | Marietta, GA May 2025 – June 2026

Vice President and Team Operations Lead *December 2025 – June 2026*

- Coordinated schedules and communication across the Rocket Design, Avionics, and Payload subteams for the Intercollegiate Rocket Engineering Competition (IREC) rocket
- Used high-fidelity CFD to optimize rocket body and camera mount geometry for reduced aerodynamic drag
- Assisted with fiberglass body construction and fixed fin-to-airframe gaps on the fin can with iterative SolidWorks fixes
- 3D printed the camera mount and payload components and standardized team build documentation
- Led the administrative team organizing public outreach events
- Organized OpenRocket, SolidWorks, and ANSYS STEM workshops, expanding team technical capabilities and outreach

Liquid Propulsion Lead *May 2025 – December 2025*

- Led propellant and injector design trade studies for a nitrous oxide/ethanol bipropellant engine using CAD and CFD
- Sized the engine for 250 lbf thrust at an O/F ratio of 4.8 from thrust equations, injector pressure drop, chamber pressure, and mass flow
- Designed an injector water cold-flow test stand on a pressurized nitrogen supply (150–220 psi)
- Refined a coaxial swirl injector assembly with 60° swirl ports through SolidWorks design checks before fabrication

Kennesaw Aerial Robotics Team (SAE Aero Design) | Marietta, GA September 2023 – April 2025

PADA Team Lead

- Led 20+ members through design, manufacturing, and flight testing of a powered autonomous delivery aircraft (PADA)

launched from a larger carbon fiber carrier aircraft, from CAD through fabrication and integration
- Placed 10th of 19 teams in the Advanced Class at the 2024 international SAE Aero Design competition
- Built a carbon fiber mold and performed a wet layup for the fuselage
- Used ArduPilot to fly a custom flying wing autonomously after mid-flight release, navigating to landing coordinates detected in flight
- Configured a Matek H743 flight computer and tuned its autopilot for stable flight on an inherently unstable airframe
- Taught a three-part ANSYS Fluent workshop series to KSU competition team members

ENGINEERING PROJECTS

Ship-Launched Multi-Domain ISR UAV – Aircraft Design and Performance (ENGR 3802) Spring 2026
- Led airfoil selection, CAD/CFD, and final sizing on a 4-person team for a tailless flying-wing UAV performing air and underwater ISR: 15.4 ft span, 40° leading-edge sweep, 2,050 lb takeoff weight, single internal PW610 turbofan
- Exceeded requirements with 1,060 NMi range (vs. 1,000), 47 min loiter (vs. 30), 89 kt stall speed (vs. 97 kt limit), and cruise L/D of 13.5 at Mach 0.7
- Designed the dual-domain transition: sealed inlet and exhaust doors, deep-stall water landing at about 10° pitch, flooding wing-root tanks as ballast, and retractable wingtip thrusters for underwater transit
- Ran a four-concept trade study (efficiency, stealth, structural complexity, stability, dual-domain capability) and specified a hybrid CFRP and aluminum 5083 structure with Inconel exhaust and EPDM seals

Tandem-Wing VTOL UAV for the DARPA Lift Challenge – Student Design Project Fall 2025 – Spring 2026
- Led aerodynamic and planform design on a 6-person team for a blown-wing VTOL concept meeting the challenge's 4:1 payload-to-weight target (under 55 lb aircraft lifting 110+ lb)
- Placed eight propellers across two tandem wings to use slipstream for added lift; built fuselage and wing sections from foam core with fiberglass reinforcement

Pumpkin Launch Trebuchet – Dynamics Fall 2024
- Designed and built a 12 ft trebuchet with a 6 ft arm and 200 lb poured-cement counterweight capable of launching 15 lb pumpkins, sizing the arm, sling, and counterweight from first-principles dynamics

Re-entry Vehicle Heat Shield Materials Review – Material Science Spring 2025
- Reviewed ceramic and metal alloy heat shields with 2 teammates, then wrote a report and presented on their application, manufacturing, and thermal performance

Gravity-Launched Vehicle – Introduction to Mechanical Engineering Spring 2024
- Engineered a self-steering vehicle guided by a wire track with consistent results, documented in a technical design report

SCHOLARSHIPS AND AWARDS

- Junior Scholars Program, KSU Office of Undergraduate Research ($2,000) August 2025
- SPCEET Faculty Scholarship Grant for transonic delta wing research under Dr. Gaurav Sharma ($3,000) Spring 2025
- Dean's List, Kennesaw State University Spring 2024, Fall 2024, Spring 2025
- First Place, KSU College of Computing and Software Engineering Ideathon ($150) 2024
- Sophomore Scholars Program, KSU Office of Undergraduate Research ($2,000) August 2024
- Summer Undergraduate Research Program (SURP), KSU Office of Undergraduate Research ($4,000) April 2024
- President's List, Kennesaw State University Fall 2023
- First-Year Scholars Program, KSU Office of Undergraduate Research ($2,000) August 2023
- Georgia HOPE Scholarship, full undergraduate tuition May 2023

PROFESSIONAL ORGANIZATIONS

- American Institute of Aeronautics and Astronautics (AIAA), Student Member October 2023 – Present
- American Society of Mechanical Engineers (ASME), Student Member February 2024 – February 2025
- American Physical Society (APS), Student Member October 2023 – October 2024

TECHNICAL SKILLS

CFD and Aerodynamics: ANSYS Fluent, OpenFOAM, FlightStream, OpenVSP, SolidWorks Flow Simulation, ParaView
Controls and Autonomy: MATLAB/Simulink, PX4, ArduPilot, Pixhawk, X-Plane
Programming: MATLAB, Python, R, LaTeX, Linux
CAD and Design: Siemens NX, SolidWorks, AutoCAD, ANSYS SpaceClaim, GD&T, Tolerance Stack-Up, DFA/DFM, Sheet Metal
Manufacturing: Composite Wet Layup, 3D Printing, Laser Cutting, Lathe/Mill, CNC Machining, Foam Cutting
Software: SharePoint, Microsoft Office, Inkscape, OriginPro